

# KAĞAN ONUR

Data Analyst • Data Scientist

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## PROFILE

Statistics graduate of Yıldız Technical University (2026). What interests me is getting from data to a decision: my thesis modelled 21 years of Turkish investment instruments to measure which one actually protects savings against inflation and currency risk; a panel data project tested whether discount campaigns genuinely change buying behaviour; and I represented Türkiye at ASA Data Quest 2025 alongside teams from four continents. I work in Python, R, Stata and SQL, and I treat explaining *why* a model behaved as it did as part of the job rather than an afterthought. Alongside this I designed, built and shipped mesite.org single-handedly. Open to data analyst and data scientist roles.

## SELECTED PROJECTS

### Time Series Analysis of Turkish Investment Instruments

Undergraduate Thesis • Yıldız Technical University • 2026

Advisor: Prof. Dr. Öyküm Esra Yiğit

Which asset actually protects savings against Türkiye's inflation and currency swings — and can deep learning forecast these markets better than a classical model? I tested both questions on 21 years of monthly data across five series.

- Gold was the only instrument that held its real value over two decades; deposit returns were eroded by inflation.
- Adding the 2008, 2018 and 2021 crises to the model improved the gold forecast fourfold ( $R^2$  0.12 → 0.51).
- Classical ARIMA beat the deep learning LSTM. I looked into why rather than taking the win: the comparison wasn't fair, because the LSTM forecasts on top of its own forecasts and the error compounds over 46 months.
- Confirmed that currency moves drive inflation rather than the reverse — a measure of exchange-rate pass-through.

Methods: ADF / PP / KPSS unit-root tests • Bai–Perron structural breaks • ARDL bounds testing • ARMA-GARCH(1,1) and EGARCH • Granger causality • ARIMA vs LSTM vs Holt benchmarking. Python (statsmodels, arch, TensorFlow/Keras).

### MeSite — Branching Video Platform

Individual project • mesite.org • 2026, ongoing

A video platform where stories branch: you watch a clip, and if you don't like where it goes you upload your own continuation. I built the whole thing alone — data model, algorithm, API, interface and server — and it is live in closed beta.

- Designed the algorithm that decides which branch becomes the main storyline. I measured attention in **capped seconds rather than completion rate**, because completion rate would push everyone to cut their videos shorter.
- Made diversity and attention **multipliers rather than added points**, so a branch inflated by one author or watched by nobody can't become the main line however large it grows.
- Set up watch-time collection so the algorithm's coefficients can be tuned against real behaviour instead of staying at estimated values.
- Shipped it end to end: nightly backups with a tested restore, rate limiting, and account deletion under Turkish data protection law.

Stack: Python (FastAPI, SQLAlchemy) • PostgreSQL for accounts, Neo4j for the branching graph • Docker • Caddy • JWT • FFmpeg transcoding • JS/HTML/CSS front end. ~8,600 lines of Python with an automated test suite, deployed on a Hetzner VPS.

### Effect of Price Discounts on Player Counts on Steam

Panel Data Analysis • Team of 3 • 2026

Do Steam sales actually bring players back? We tested it on twelve months of data from nine popular games.

- Price matters: a **1% increase costs 0.57% of the player base**.
- The sales themselves did not. None of the four seasonal sale periods produced a significant lift.
- The reason was a measurement trap: **98% of the difference in player numbers comes from how big the game already is**. Discounts don't grow a game — large games simply stay crowded.

Methods: balanced panel of 108 observations (9 games × 12 months) • a combined "discount pressure" variable merging depth and duration • panel unit-root tests • Hausman test selecting fixed effects • game-clustered robust standard errors. Stata, R.

## Data Science & Sustainable Development

### ASA Data Quest 2025 · International · Representing Türkiye

American Statistical Association (CIRS) · Mentor: Ceylan Yozgatligil · Team of 4

Do countries have to accept environmental cost in order to develop? We presented Türkiye's answer against teams from Brazil, Indonesia, India, Kenya and China.

- Wealth, education and life expectancy rise together almost in lockstep (**0.87–0.92**).
- Carbon cost is where they part: national income tracks emissions closely (**0.87**) while education and health spending do not — social investment develops a country at a lower carbon price.
- A model weighing economy, education, health and emissions together put the Nordic countries and Costa Rica on top, showing the trade-off isn't inevitable.

Validation: Durbin–Watson, VIF and Shapiro–Wilk diagnostics.

## Customer Purchasing Behaviour Analysis

### Individual project · 2026

Tested what drives a customer's decision to buy — age, gender and review quality — on an open-access dataset. Gender was the only statistically significant driver.

Methods: multivariate logistic regression · Pearson chi-square tests.

## EXPERIENCE

### Data Annotation Specialist

Jan 2026 – Feb 2026

#### Artiwise — AI-Powered Software Company

- Labelled and classified text data for supervised NLP models at a company building AI-powered document analysis for customer-centric process management.
- Applied and refined annotation guidelines to keep labels consistent, and ran quality checks on completed batches to keep training data reliable.

## EDUCATION

### Yıldız Technical University

2022 – 2026

#### BSc Statistics, Faculty of Arts and Sciences

Relevant coursework: Econometrics, Time Series Analysis, Panel Data Analysis, Multivariate Statistics, Regression Analysis, Data Visualization, Data Management

### İstanbul Kültür University

2020 – 2022

#### Computer Engineering, Faculty of Engineering

English preparatory programme completed; transferred to Statistics at YTU

### Gürlek Nakipoğlu Anatolian High School

2015 – 2019

## TECHNICAL SKILLS

|                                       |                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; Tools</b>        | Python (pandas, NumPy, statsmodels, scikit-learn, TensorFlow/Keras) · R · Stata · SQL · Excel · SPSS                    |
| <b>Time Series &amp; Econometrics</b> | ARIMA / SARIMA · ARMA-GARCH · EGARCH · ARDL bounds testing · Granger causality · Bai–Perron · ADF / PP / KPSS           |
| <b>Panel &amp; Regression</b>         | Fixed and random effects · Hausman test · cluster-robust standard errors · panel unit roots · OLS · logistic regression |
| <b>Databases &amp; Engineering</b>    | PostgreSQL · Neo4j (graph database) · FastAPI · Docker · Git · REST API design                                          |
| <b>Machine Learning</b>               | LSTM and deep sequence models · supervised classification · model benchmarking (MAE, MAPE, $R^2$ )                      |
| <b>Data Work</b>                      | Cleaning and merging multi-source data · feature engineering · data labelling · visualization and reporting             |
| <b>Languages</b>                      | Turkish — Native · English — Upper-Intermediate (B2), DİLKO in partnership with Pearson (2022)                          |